

[PZoom Deep Dive 9.1] Prove that $\psi(a)$ escapes the image of ϕ

Let $x \in S$. We analyze the placement of $\phi(x)$ relative to $\psi(a)$ by partitioning the domain S around the minimal element a :

- **Case 1** ($x < a$): The minimality of a ensures that $\phi(x) = \psi(x)$. Since ψ strictly preserves order, $x < a \implies \psi(x) <' \psi(a)$. Substituting $\phi(x)$ for $\psi(x)$, we obtain:

$$\phi(x) <' \psi(a) \implies \phi(x) \neq \psi(a)$$

- **Case 2** ($x = a$): By our initial asymmetric assumption, we have directly that $\phi(a) >' \psi(a)$.
- **Case 3** ($a < x$): Since ϕ strictly preserves order, $a < x \implies \phi(a) <' \phi(x)$. Combining this with our assumption $\psi(a) <' \phi(a)$ yields $\psi(a) <' \phi(x)$ by transitivity of the order in S' .

Since x was arbitrary, $\phi(x) \neq \psi(a)$ for all $x \in S$.

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